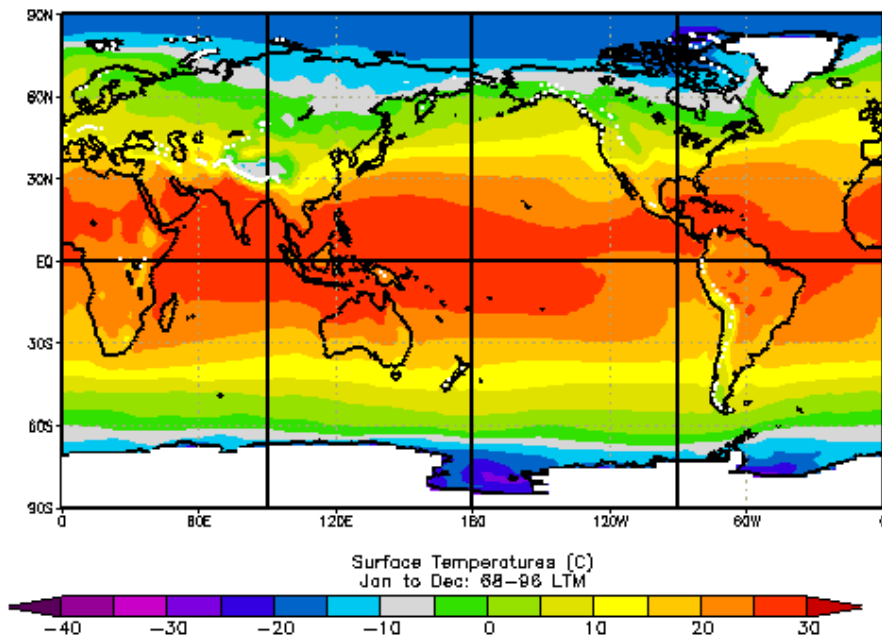
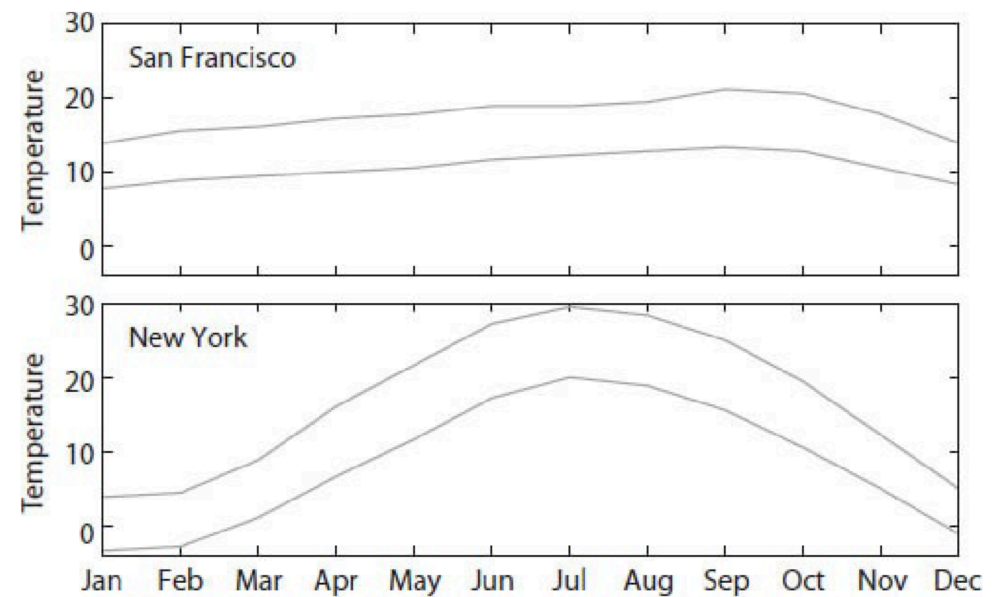


Oceans and Climate

What determines climate?

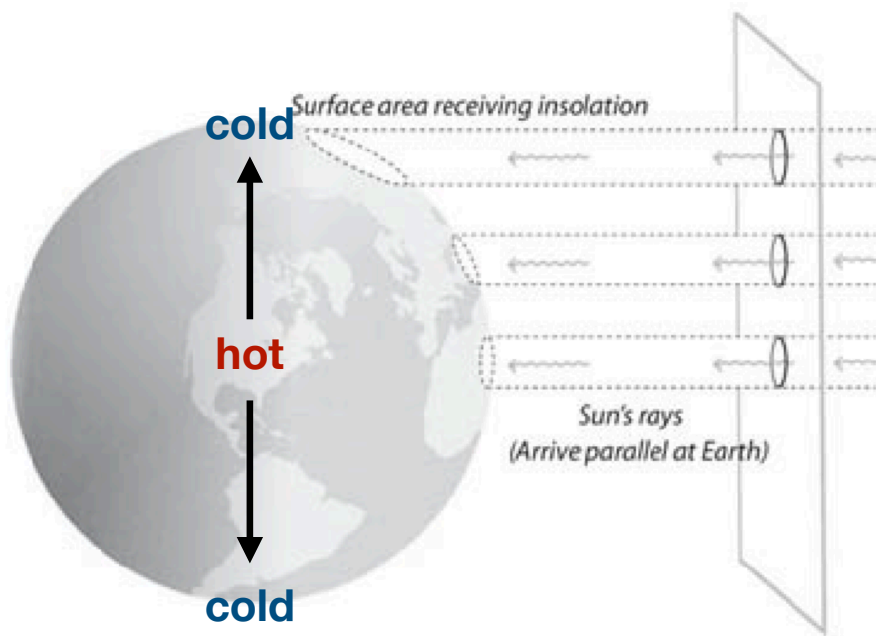


[Nullet, U Hawaii]

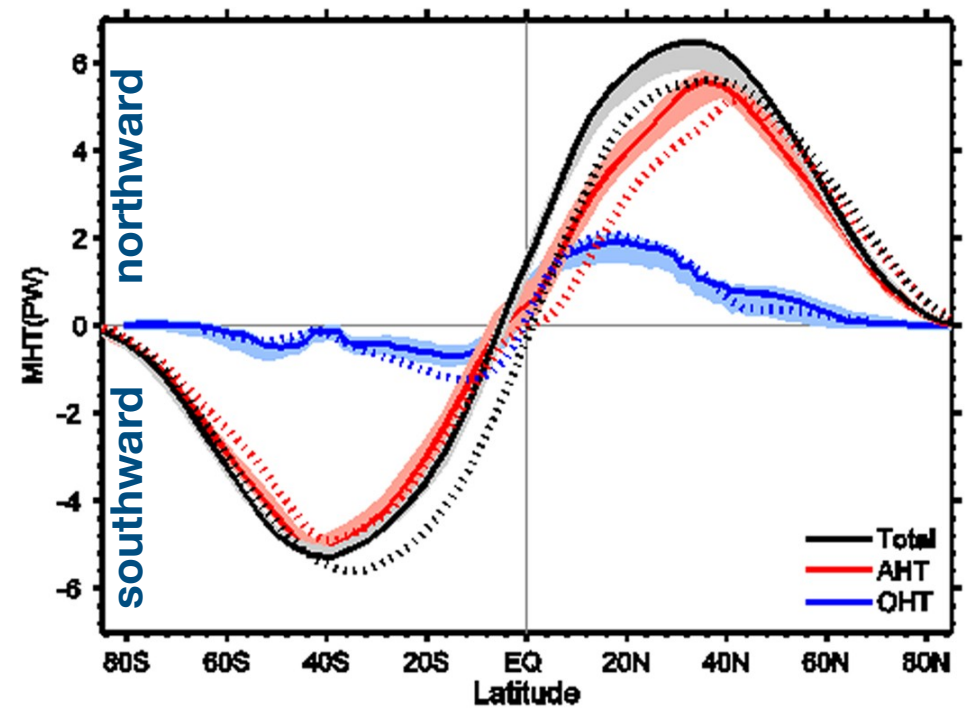


[Vallis 2011]

Ocean redistributes heat meridionally (north-south)

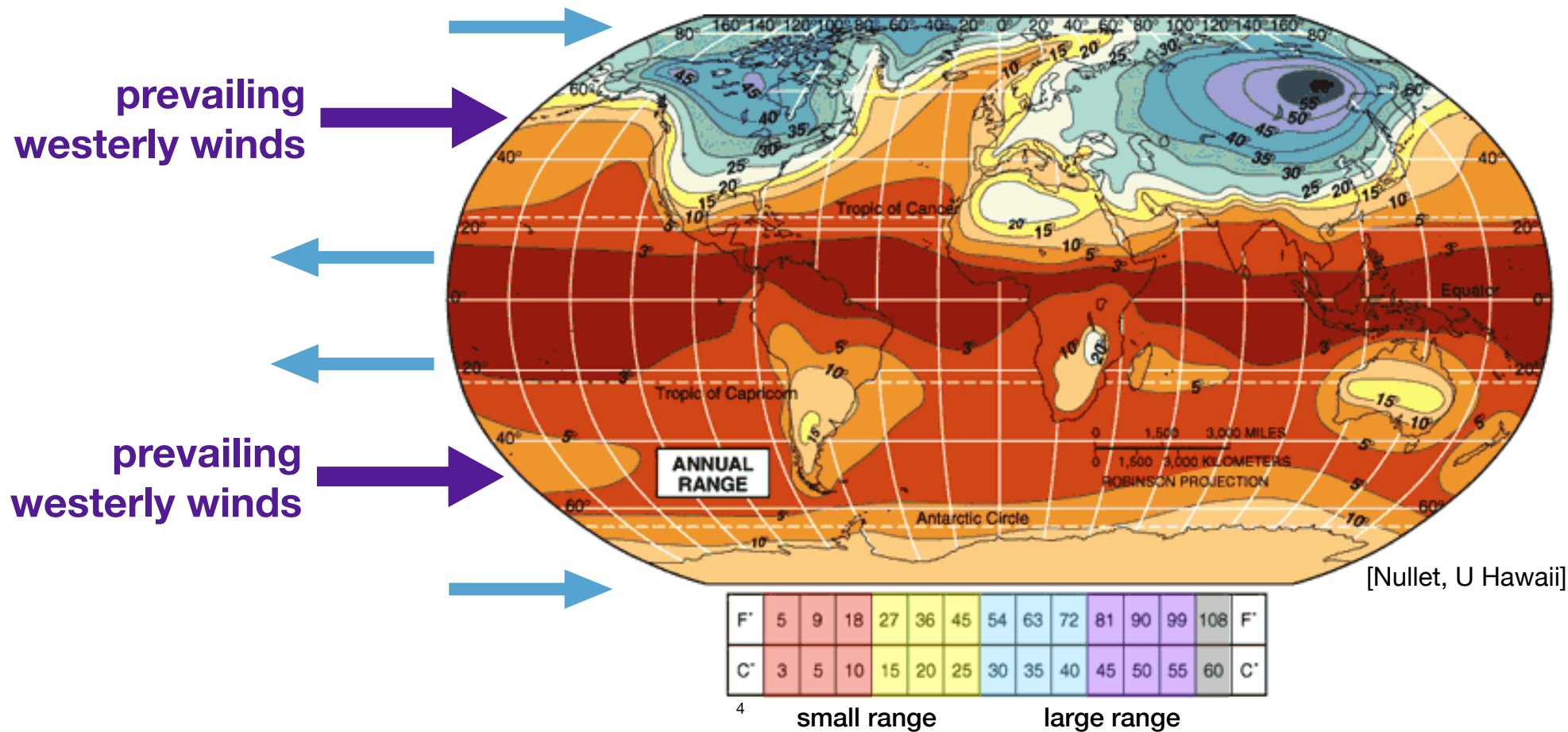


[Vallis 2011]

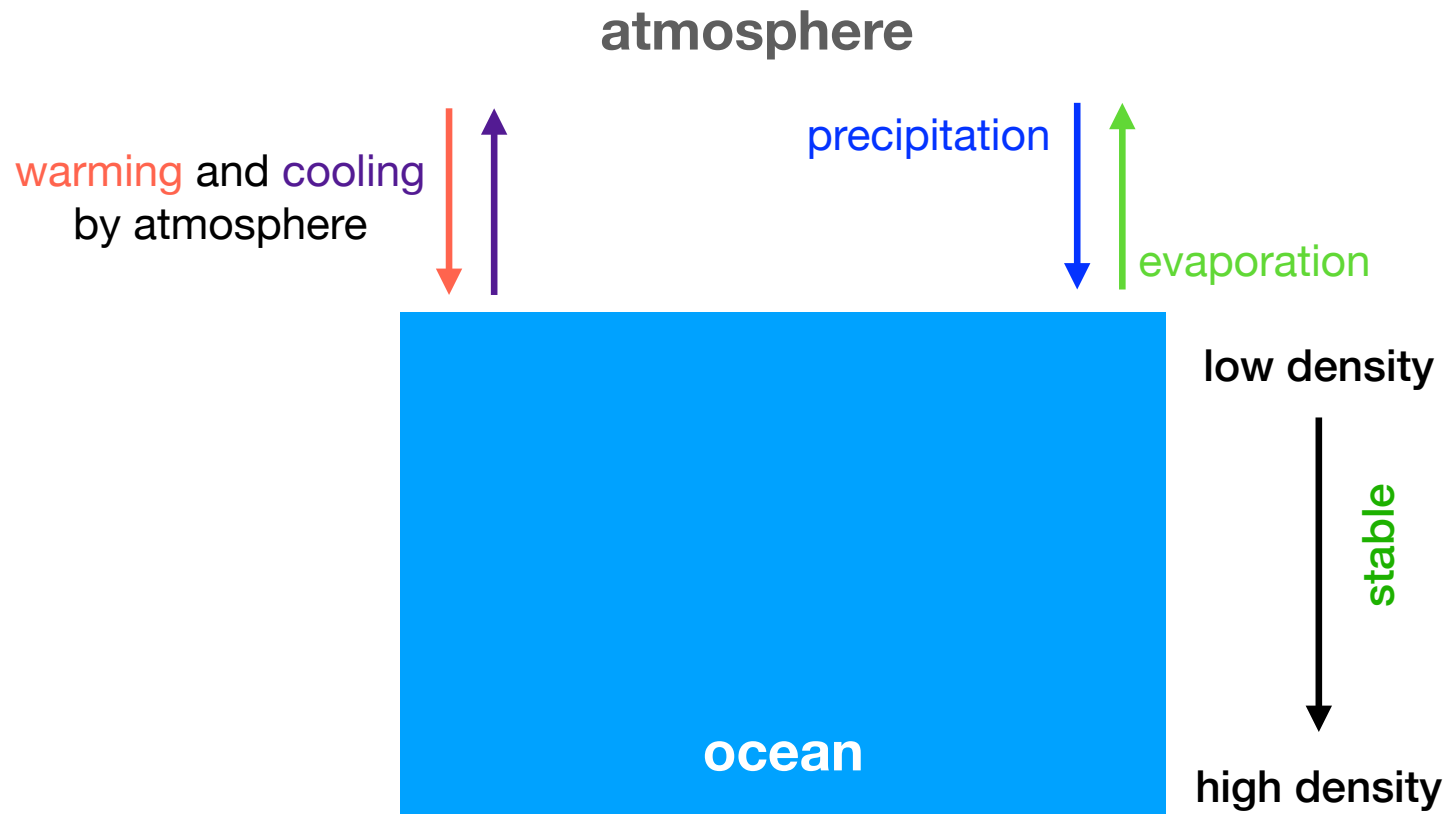


[Yang et al. 2015]

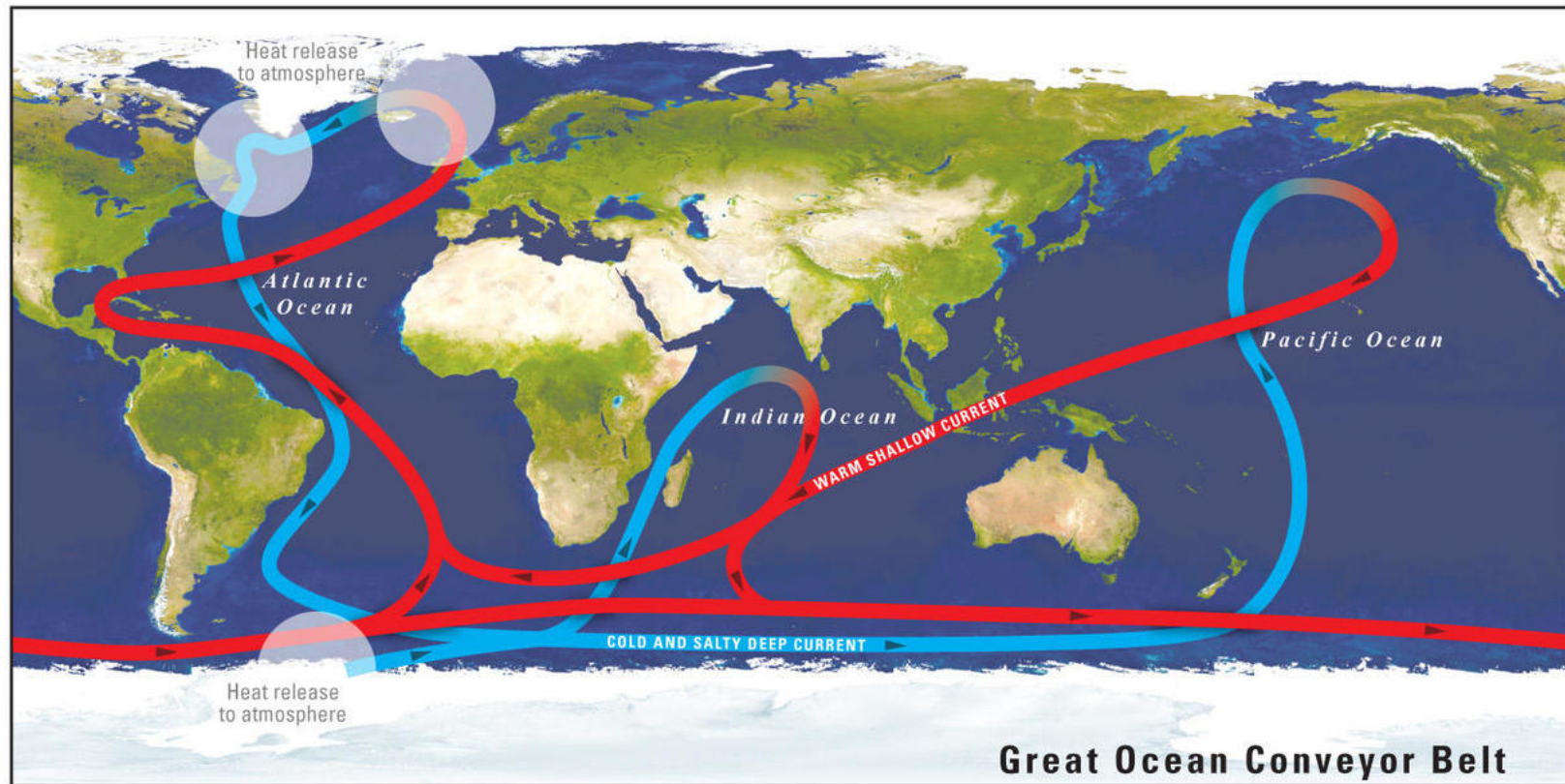
Ocean controls zonal (east-west) climate variability



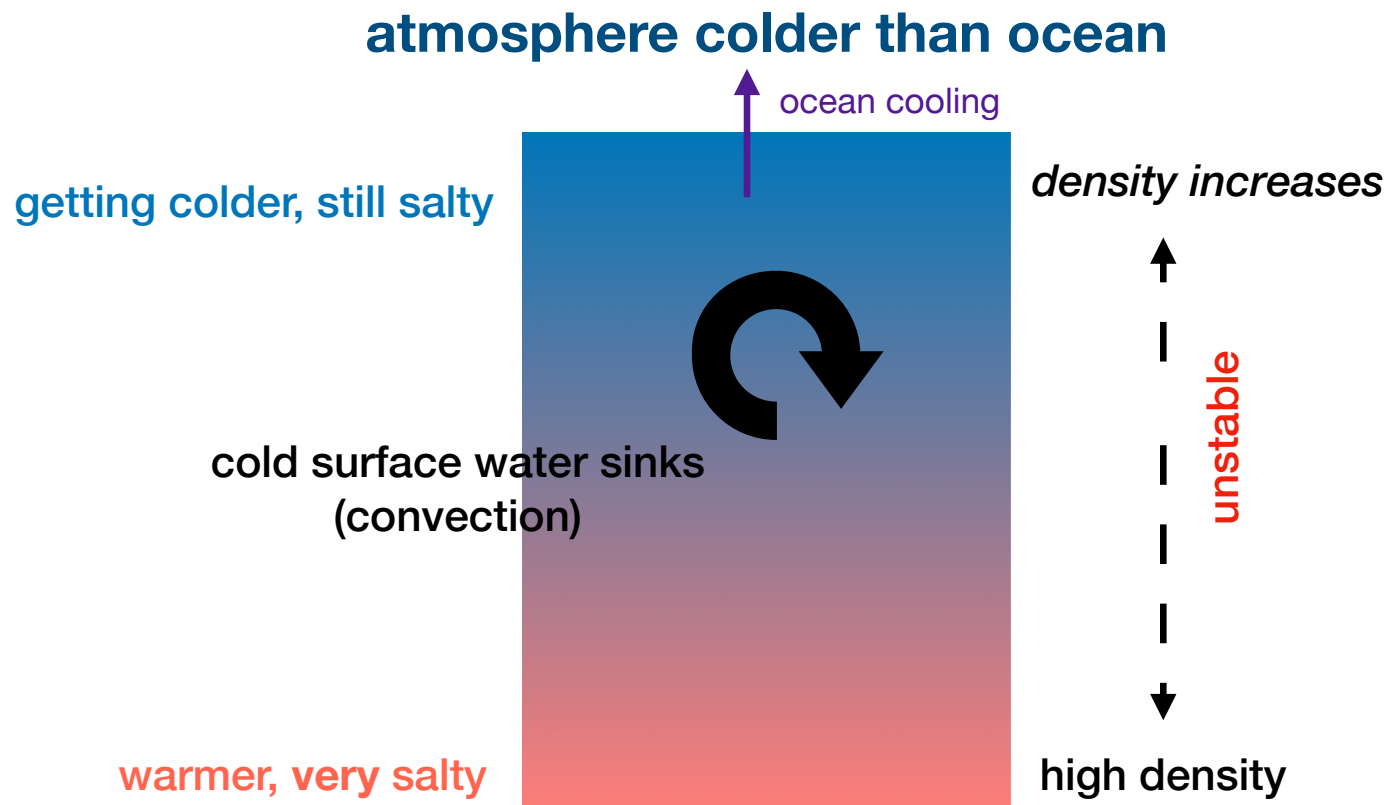
Surface heat and freshwater exchanges



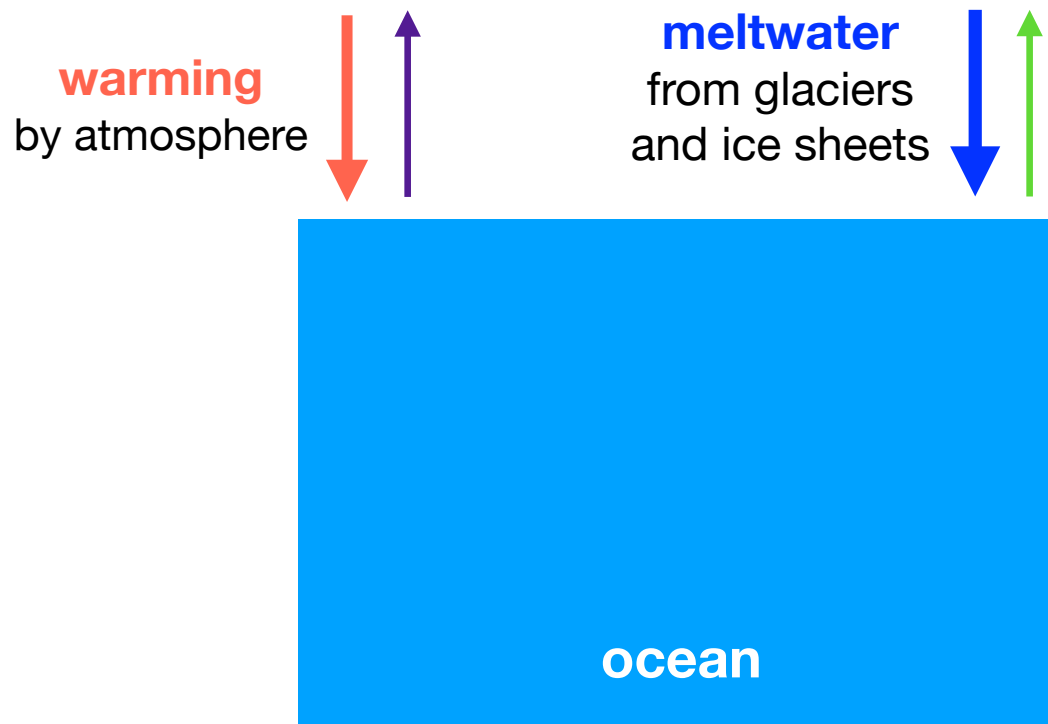
Ocean currents transport heat



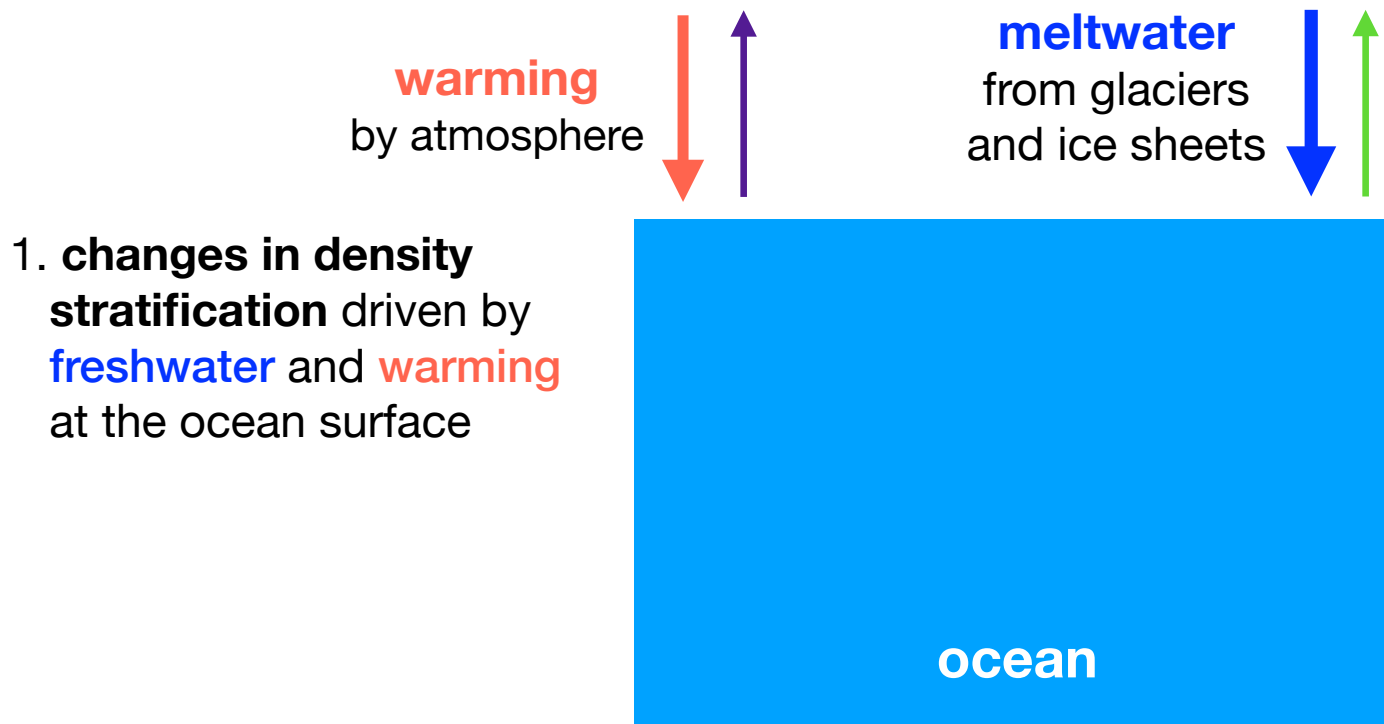
High latitude ocean convection



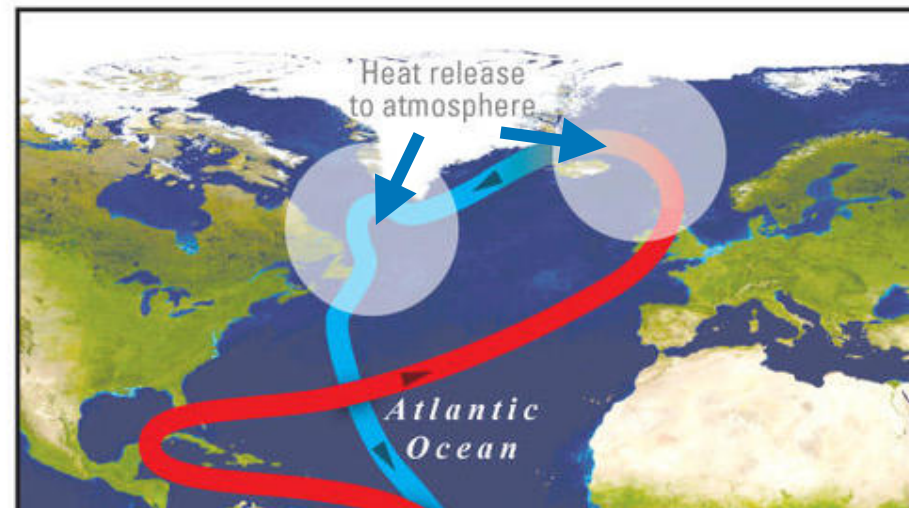
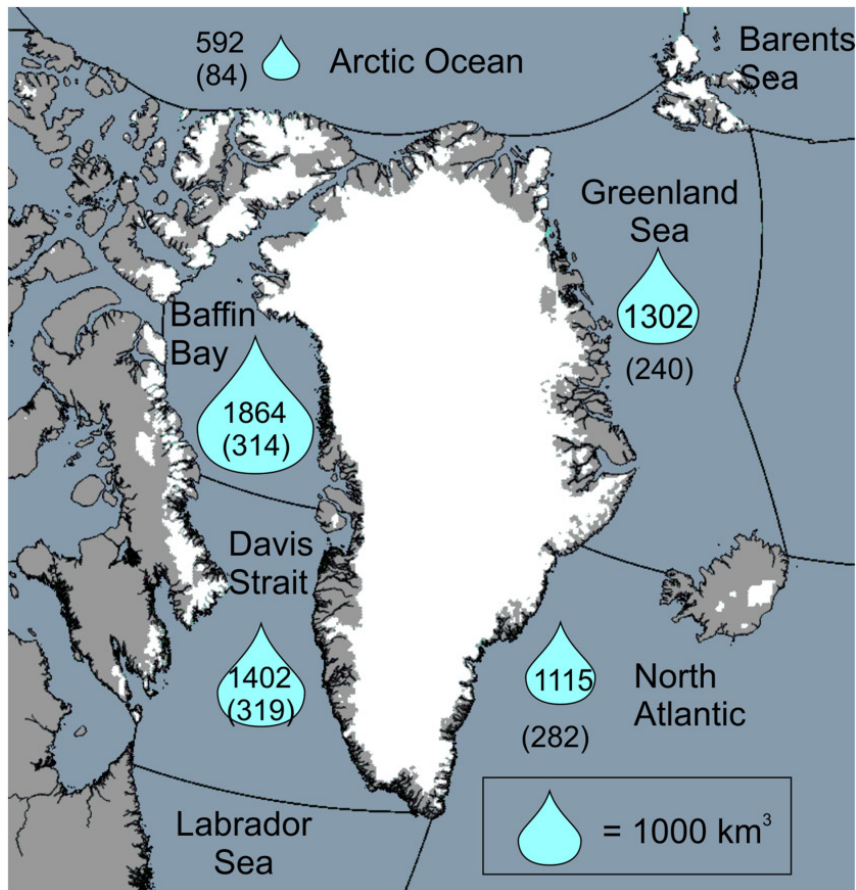
Climate change effects on the ocean



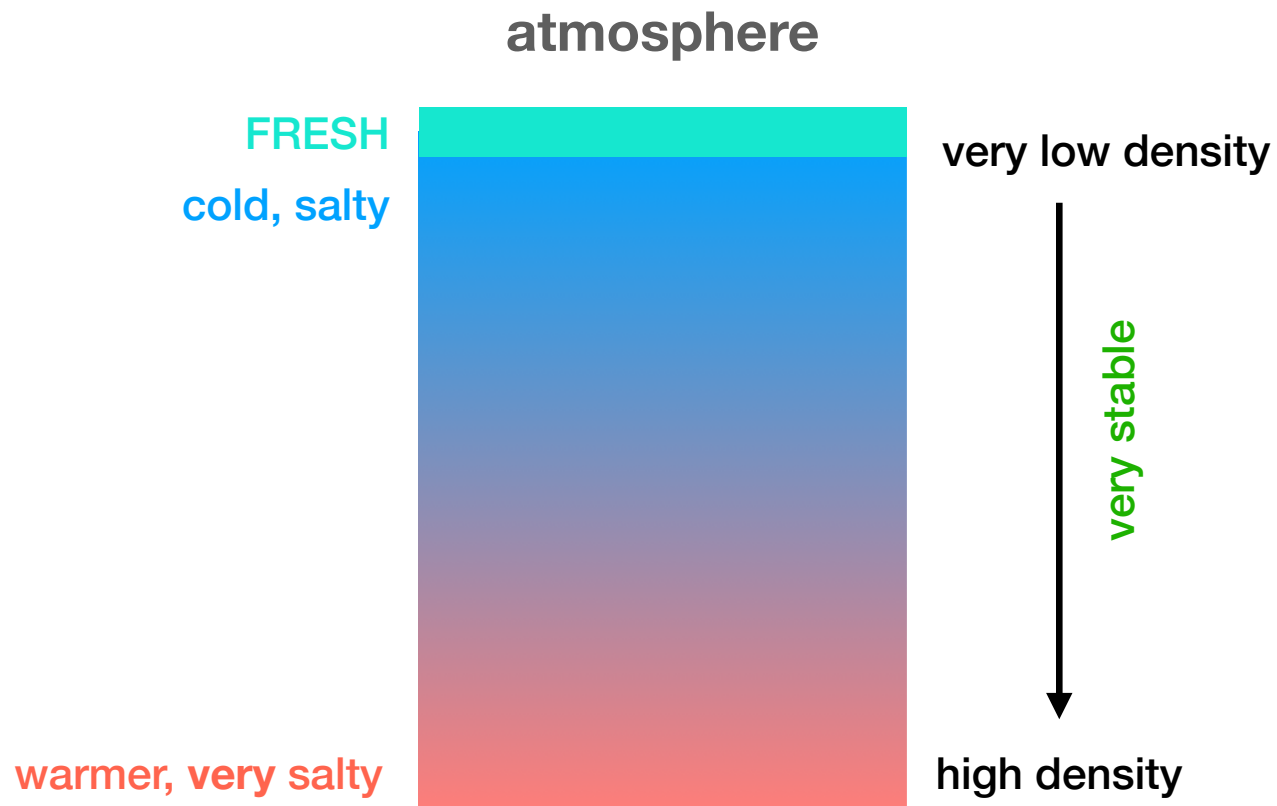
Climate change effects on the ocean



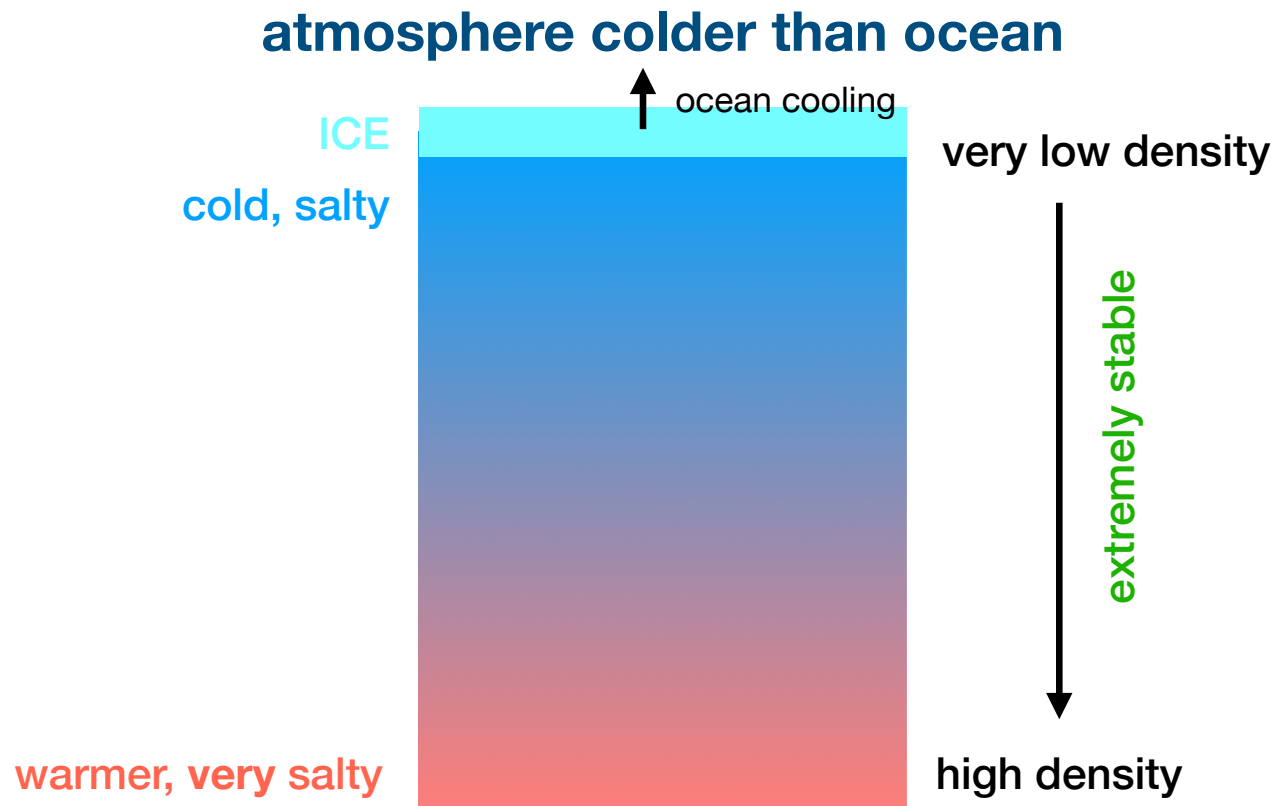
Meltwater input in convection regions



Surface freshening stabilizes density gradient



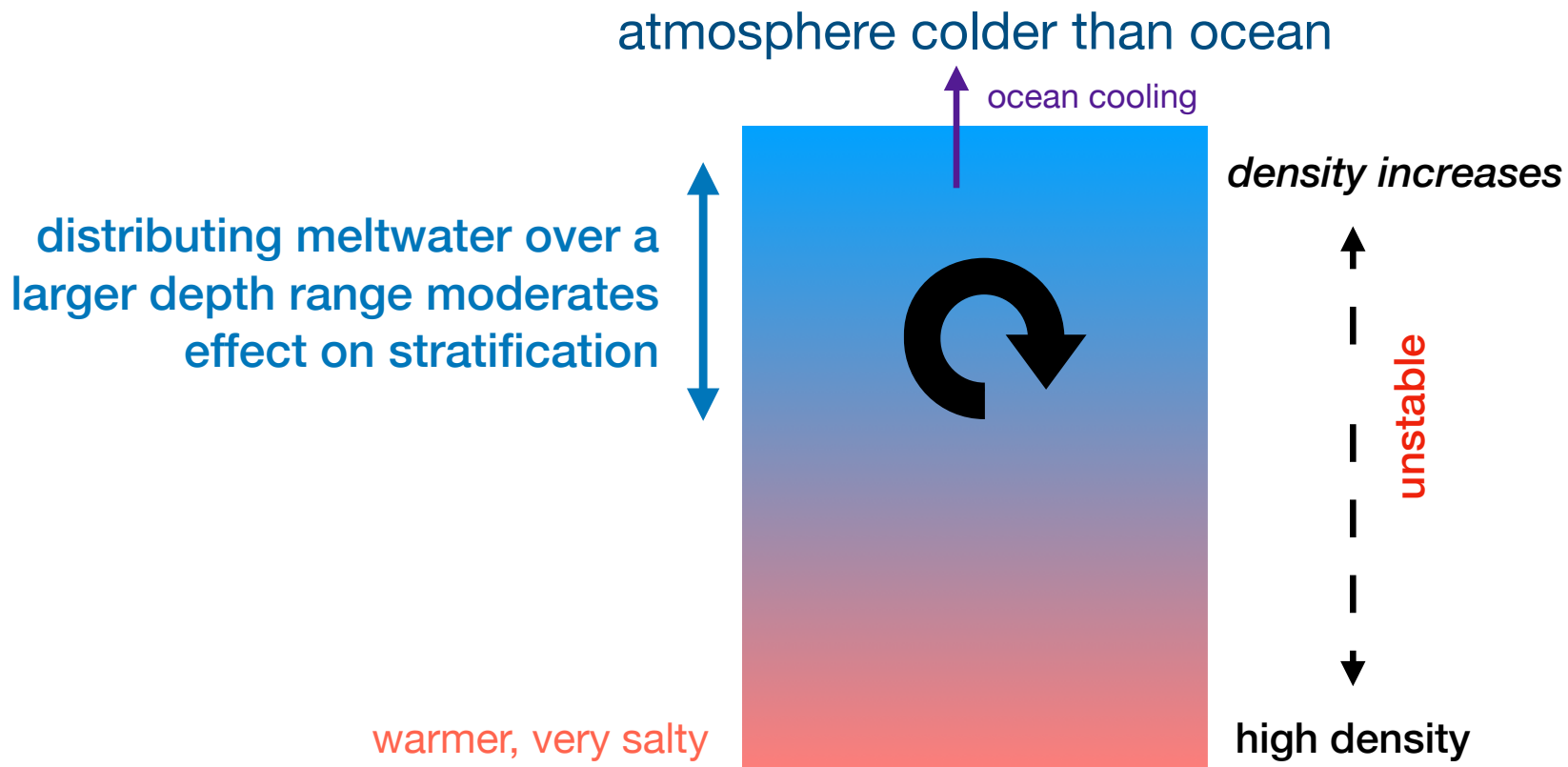
Surface freshening may inhibit convection



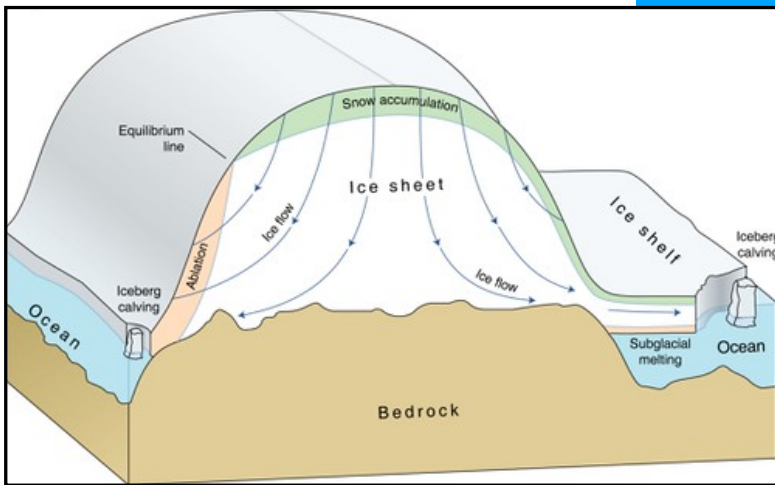
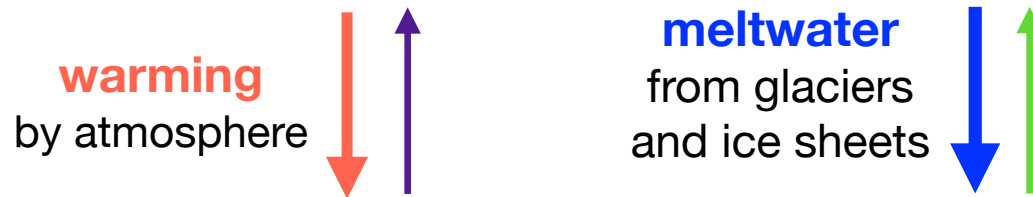
Where does meltwater end up in the ocean?



Meltwater depth matters for convection



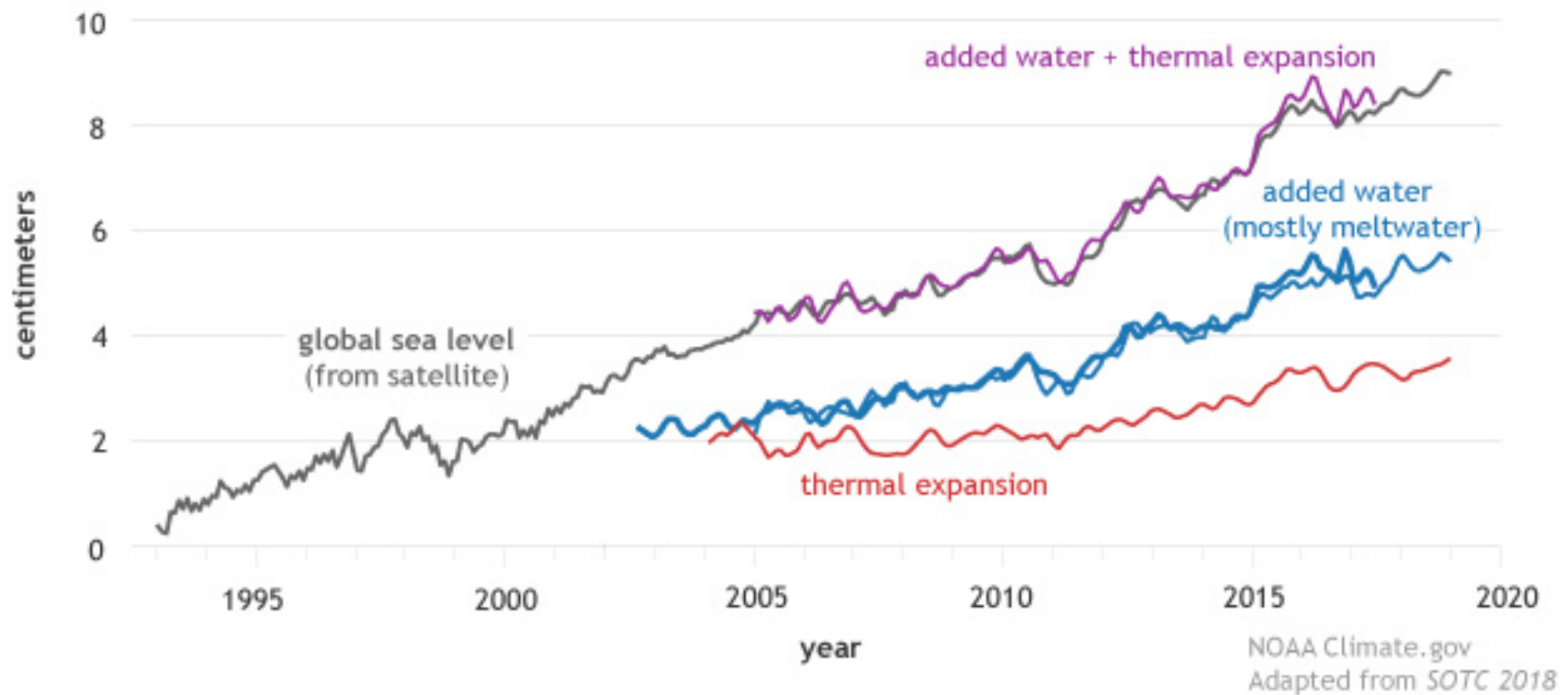
Climate change effects on the ocean



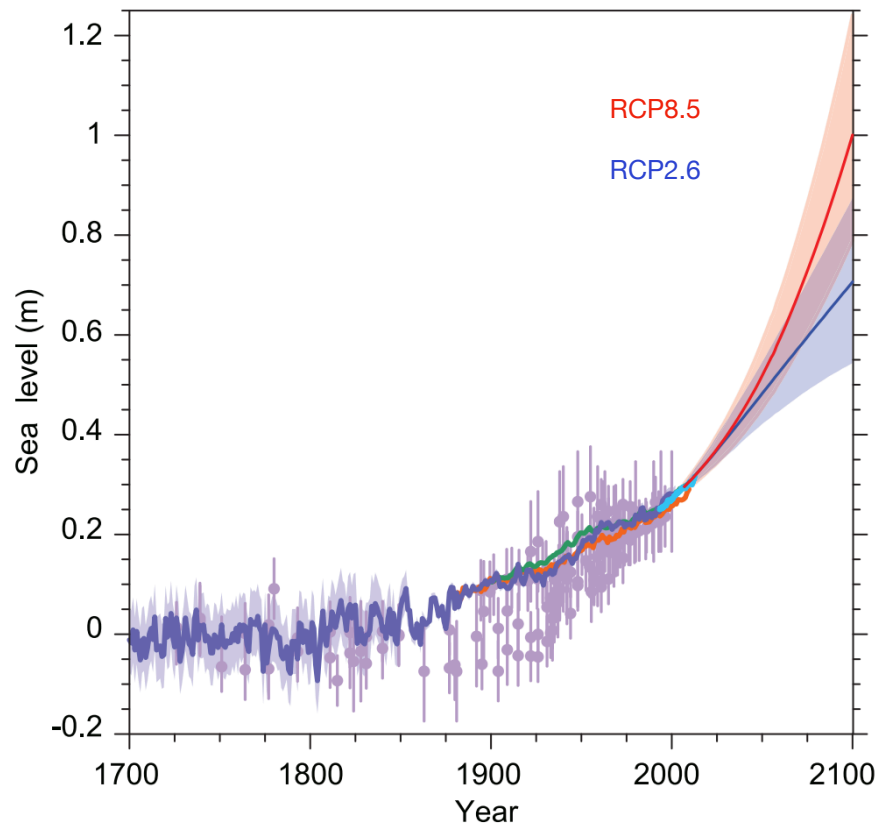
ocean

2. **sea level rise**
driven by **added mass**
and **thermal expansion**

Contributors to sea level rise



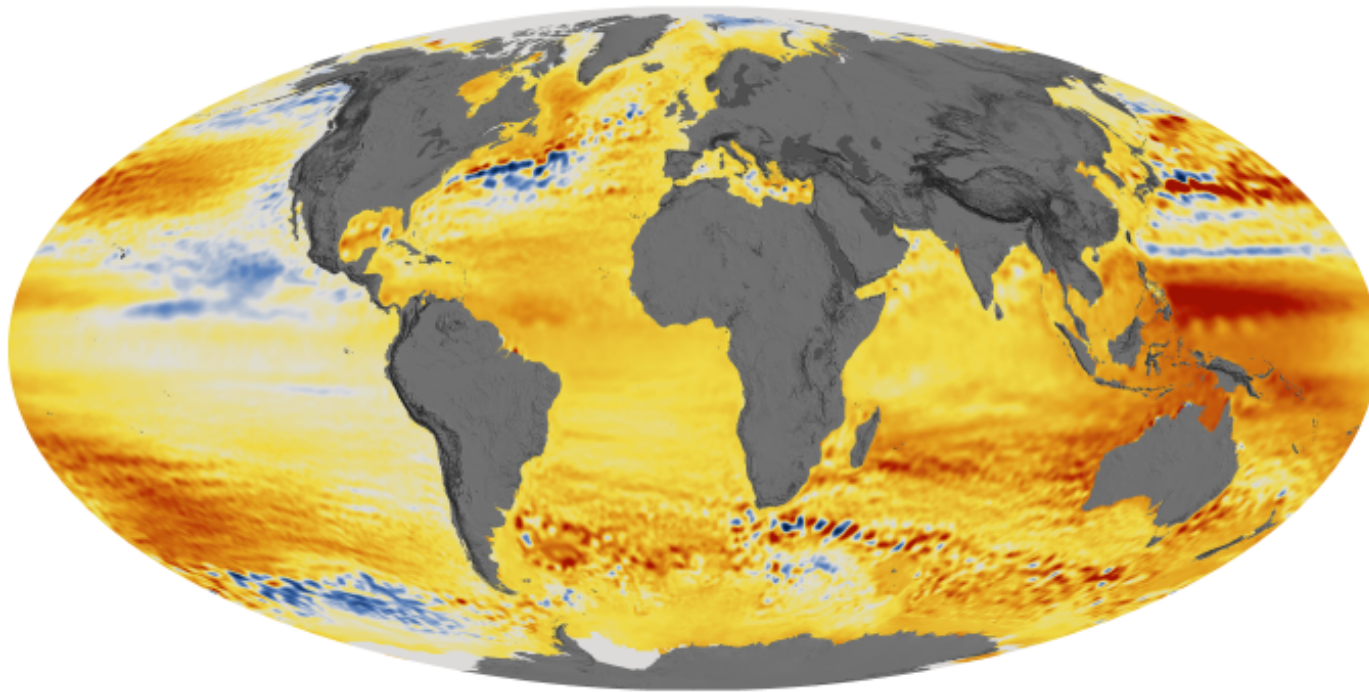
Uncertainty in future sea level rise



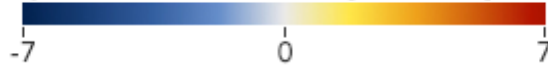
This is global mean sea level.

What about local adaptation?

Regional variability in sea level rise



22-year Sea Surface Height Change (cm)



Local variation in sea level

